4 **(a)** Define 'impulse' in words.

..... [1]

(b) A student jumps from the top of a low wall to the ground. The mass of the student is 55 kg. As the student reaches the ground, her speed is 4.6 m / s.

(i) The student stops moving 0.95 s after her feet first touch the ground.

Calculate the resultant force exerted on the student as she stops moving.

resultant force = ... N [2]

(ii) As the student lands, she bends her knees.

Explain how this prevents the student from being injured.

Use ideas about force, momentum and time in your answer.

.....

.....

..... [3]

(iii) Another student stands on the ground.

The area in contact with the ground is 380 cm^2.

The weight of the student is 530 N.

Calculate the pressure exerted by the student on the ground. State the unit.

pressure = ... unit = ... [2]

[Total: 8]

© Cambridge University Press & Assessment 2026

DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN

PapaCambridge · papacambridge.com 0625/42 Question Paper June 2026